



PROJECT PROPOSAL

A Comprehensive Technical & Business Proposal

Project Title	NeuralQuant — LSTM Cryptocurrency Price Prediction Dashboard
Version	Bidirectional LSTM with Multi-Feature Engineering
Submitted By	FA24-BSE-078 FA24-BSE-081 FA24-BSE-090
Technology Stack	Python 3.12, TensorFlow/Keras, Streamlit, yFinance, scikit-learn
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EXECUTIVE SUMMARY

NeuralQuant is a full-stack, AI-powered cryptocurrency analytics and prediction platform built with Python and Streamlit. It harnesses Bidirectional LSTM (Long Short-Term Memory) deep learning networks — trained on 16 engineered technical features — to forecast next-day cryptocurrency closing prices. The platform combines real-time market data ingestion, RSI-based trading signals, backtested model validation, and a modern dark/light-themed dashboard UI, delivering institutional-grade predictive intelligence accessible to any authenticated user.

1. Introduction & Project Overview

1.1 Background

The cryptocurrency market is one of the most volatile and data-rich financial markets in the world, operating 24 hours a day, 7 days a week across global exchanges. Traditional analysis tools and simple statistical models have proven inadequate in capturing the complex, non-linear dynamics that drive price movements in assets like Bitcoin (BTC), Ethereum (ETH), and Dogecoin (DOGE).

Advances in deep learning, specifically Recurrent Neural Networks (RNNs) and their improved variant — Long Short-Term Memory (LSTM) networks — have opened new avenues for sequential time-series prediction. LSTMs are uniquely suited to this domain because they can retain and leverage long-range temporal dependencies within market data, making them significantly more capable than traditional ARIMA or regression-based forecasting approaches.

1.2 Project Vision

NeuralQuant was conceived to bridge the gap between cutting-edge deep learning research and practical, accessible trading intelligence. The platform aims to:

- Provide real-time, AI-generated price forecasts for any cryptocurrency listed on Yahoo Finance.
- Deliver technical analysis indicators (RSI, MACD, Bollinger Bands) in a visually rich, intuitive interface.
- Enable users to validate model accuracy through rigorous out-of-sample backtesting.
- Maintain a persistent, personalized prediction history with automatic actual-price reconciliation.

1.3 Target Audience

NeuralQuant is designed for three primary user groups:

- Retail Crypto Investors seeking AI-augmented insights beyond standard exchange dashboards.
- Academic Researchers exploring LSTM performance in high-volatility financial time series.
- Data Science Practitioners and students who wish to study or extend a production-quality deep learning pipeline.

2. Problem Statement

2.1 The Challenge of Crypto Price Prediction

Cryptocurrency prices are influenced by a complex, interconnected web of factors including macroeconomic events, regulatory announcements, social media sentiment, whale wallet movements, and technical chart patterns. This multidimensional nature renders simple forecasting models ineffective.

2.2 Limitations of Existing Solutions

Most existing retail-focused crypto tools fall into one of two failure modes:

1. **Oversimplification:** Tools that display lagging indicators (e.g., simple moving averages) without any predictive modelling capability.
2. **Inaccessibility:** Research-grade LSTM implementations locked behind complex Jupyter Notebooks requiring manual data engineering and no user interface.
3. **Data Leakage:** Many publicly available LSTM stock/crypto prediction tutorials train and validate on the same scaled data, producing artificially inflated accuracy metrics that do not reflect real-world performance.

2.3 The NeuralQuant Solution

NeuralQuant addresses all three limitations by delivering a fully engineered, leakage-free LSTM pipeline packaged within a professional, authenticated web dashboard. The model is trained exclusively on historical data and validated on held-out future data, ensuring that reported MAPE scores reflect genuine predictive performance.

3. System Architecture

3.1 Architectural Overview

NeuralQuant follows a clean two-layer architecture separating backend logic from frontend presentation:

Layer	Description
<code>dashboard.py</code>	Backend engine: all data fetching, feature engineering, LSTM model building, training, prediction, backtesting, and user management. Zero UI code — pure Python logic.
<code>frontend.py</code>	Streamlit-based frontend: session state management, CSS theming, component rendering, user interaction handling. Imports exclusively from <code>dashboard.py</code> .
<code>app.py</code>	Standalone LSTM prototype for Apple (AAPL) stock — the foundational proof-of-concept that preceded the full NeuralQuant platform.
<code>CryptocurrencyData.csv</code>	Master coin directory: 4,150 rows × 12 columns including rank, price, 24h/7d/30d change, volume, supply, and market cap.
<code>portfolio_data.csv</code>	Historical portfolio pricing data for AMZN, DPZ, BTC, NFLX — 1,520 rows of daily close prices for analysis and benchmarking.

`*_prediction_history.txt`

Per-user prediction logs: timestamped coin, predicted price, and resolved actual price — updated automatically on next login.

3.2 Data Flow Pipeline

The end-to-end data flow from raw market data to displayed prediction follows these stages:

4. User Authentication — login verified against `users.txt`; session state initialized.
5. Coin Selection — user selects any coin from the 4,150-row master CSV directory.
6. Live Data Fetch — yfinance downloads OHLCV data from 2020-01-01 to present; MultiIndex columns flattened.
7. Feature Engineering — 16 technical features computed (RSI, MACD, Bollinger Bands, EMAs, Volume SMA, momentum returns).
8. Data Scaling — MinMaxScaler fitted on training rows only (no leakage into test set).
9. Sequence Construction — 60-day rolling windows built for both X (features) and y (next Close).
10. Model Training — Bidirectional LSTM trained with EarlyStopping; typically converges in 15–30 epochs.
11. Prediction / Validation — next-day price predicted or 20-day backtest executed.
12. History Logging — result appended to user's prediction file with actual price resolved on next session.

4. Machine Learning Model — Bidirectional LSTM

4.1 Why LSTM?

Standard feedforward neural networks process inputs independently, making them blind to temporal context. LSTM networks address this through a gated memory cell architecture containing three learnable gates:

- Forget Gate: Determines what fraction of the previous cell state to discard.
- Input Gate: Controls what new information from the current time step enters the cell state.
- Output Gate: Governs what portion of the cell state is exposed as the hidden state output.

This architecture enables LSTMs to model dependencies spanning hundreds of time steps — critical when, for example, a price pattern from 45 days ago is relevant to today's prediction.

4.2 Bidirectional Architecture

NeuralQuant v2.4 extends the standard LSTM with a Bidirectional wrapper on the first layer. Bidirectional LSTMs process the input sequence in both forward and backward directions, doubling the contextual information available at each time step. While in live prediction mode (forecasting tomorrow's price) only the forward pass is truly causal, the bidirectional layer meaningfully improves pattern recognition during training by exposing the model to richer temporal context.

4.3 Model Architecture Details

Layer	Type / Units	Dropout	Return Sequences
Layer 1	BiLSTM — 128 units	30%	True
Layer 2	LSTM — 64 units	20%	True
Layer 3	LSTM — 32 units	20%	False
Layer 4	Dense — 16 neurons	—	—
Output	Dense — 1 neuron	—	—

Optimizer: Adam | Loss: Mean Squared Error | Lookback Window: 60 days | Batch Size: 32 | Max Epochs: 50 (prediction) / 30 (validation)

4.4 The 16 Engineered Features

A core design decision in NeuralQuant v2.4 was to move beyond single-feature (Close price only) inputs toward a rich 16-feature representation. The following table summarizes each feature and its financial significance:

#	Feature	Description & Significance
1–5	Open, High, Low, Close, Volume	Raw OHLCV — the five foundational market data points sourced directly from Yahoo Finance.
6	RSI (14)	Relative Strength Index: measures momentum by comparing average gains to losses over 14 periods. Values below 30 signal oversold (BUY), above 70 signal overbought (SELL).
7	MACD	Moving Average Convergence Divergence (12/26 EMA difference): captures trend direction and momentum shifts at medium-term horizons.
8	MACD Signal	9-period EMA of MACD line. MACD crossing above Signal generates bullish signal; crossing below is bearish.
9	MACD Histogram	Difference between MACD and Signal. Provides early warning of momentum changes before crossovers occur.
10	BB Upper	Upper Bollinger Band ($SMA_{20} + 2 \times STD_{20}$). Price touching the upper band indicates potential overextension.

11	BB Lower	Lower Bollinger Band ($SMA_{20} - 2 \times STD_{20}$). Price touching the lower band indicates potential undervaluation.
12	BB Width	Normalized band width $((Upper - Lower) / SMA_{20})$. Measures market volatility — narrow bands precede large breakout moves.
13	EMA 10	10-day Exponential Moving Average. Captures short-term trend with more weight on recent prices than a simple MA.
14	EMA 50	50-day Exponential Moving Average. Long-term trend indicator; EMA10 crossing EMA50 generates golden/death cross signals.
15	Vol SMA	10-day Volume Simple Moving Average. Contextualizes price moves — high volume confirms trend; low volume suggests weak conviction.
16	Return 1d / 5d	1-day and 5-day percentage price changes. Encode short and medium-term momentum directly into the feature space.

4.5 Anti-Leakage Validation Protocol

A critical bug in many publicly available LSTM tutorials is data leakage: the `MinMaxScaler` is fitted on the entire dataset (including test rows) before splitting. This artificially inflates validation accuracy because the model has implicitly "seen" the test data through the scaler's min/max parameters.

NeuralQuant's `run_validation()` function corrects this by:

13. Splitting the dataset into training rows and test rows before any scaling.
14. Fitting the `MinMaxScaler` exclusively on training rows.
15. Applying `scaler.transform()` (not `fit_transform()`) to test rows.
16. Running the 20-day backtest on truly unseen, independently scaled data.

This ensures that reported MAPE scores reflect genuine out-of-sample predictive performance.

5. Platform Features & Functionality

5.1 Authentication System

NeuralQuant implements a file-based authentication system stored in `users.txt` using a pipe-delimited `Username|Password` format. While intentionally simple for the current prototype phase, it supports:

- New user registration with duplicate-username validation.
- Session-based login state managed through Streamlit's `st.session_state`.

- Automatic actual-price reconciliation on login: any predictions marked 'Pending' are resolved against the latest Yahoo Finance close price.

Production deployment would replace this with a hashed-password database (e.g., bcrypt + PostgreSQL) and optionally OAuth2 / SSO integration.

5.2 Coin Directory & Live Data

The master CryptocurrencyData.csv contains 4,150 cryptocurrency entries with rank, symbol, price, and change percentages. Users can search and select any coin, which triggers a live yfinance fetch from 2020-01-01 to the current date. The fetch pipeline includes MultiIndex column flattening (a known yfinance v0.2+ compatibility issue) and Series-level squeeze() operations to prevent DataFrame assignment errors.

5.3 RSI Signal Engine

The platform computes a 14-period RSI in real time for any selected coin and displays one of three actionable signals:

- BUY (RSI < 30): Asset is oversold — potential reversal buying opportunity.
- SELL (RSI > 70): Asset is overbought — potential profit-taking or short entry.
- HOLD ($30 \leq \text{RSI} \leq 70$): Neutral zone — no high-conviction signal.

5.4 Top Market Movers

The dashboard homepage displays the Top 5 Gainers and Top 5 Losers from the master CSV ranked by 24h percentage change. This gives users an immediate snapshot of the day's most significant market movements without requiring any input.

5.5 LSTM Price Prediction

Clicking 'Predict Tomorrow' triggers the full run_prediction() pipeline: feature engineering, sequence construction, model training with EarlyStopping (up to 50 epochs), and next-day close price prediction. The result is displayed with a delta versus the current price and automatically logged to the user's prediction history.

5.6 Backtest Validation

The 'Validate Accuracy' function runs run_validation(): a 20-day out-of-sample backtest that trains on all historical data except the last 20 days, then compares predictions to actual prices. The Mean Absolute Percentage Error (MAPE) is displayed with color-coded interpretation:

- MAPE < 5%: High accuracy (green) — model is performing well.
- MAPE 5–10%: Moderate deviation (amber) — usable but with caution.
- MAPE > 10%: High deviation (red) — consider retraining or using a different coin.

5.7 Prediction History

Every prediction is timestamped and persisted to a user-specific text file (e.g., jutt_prediction_history.txt). On the next login, the system automatically fetches the actual

closing price for each 'Pending' entry and reconciles the record. This creates a transparent, auditable trail of model performance over time.

Based on existing data from the jutt user, the model has achieved predictions in the range of \$78,465 — \$81,488 for BTC, with actual prices of \$80,119 — \$80,754, demonstrating solid real-world accuracy for a 1-epoch training window.

5.8 UI Design & Theming

The frontend is built with Streamlit and enhanced with 600+ lines of custom CSS implementing a professional dark/light dual-theme. Design highlights include:

- Sticky topbar navigation with live market status indicator.
- Hero section with animated beam, halo pulse, and gradient overlays.
- Lightning bolt card animations on the Top Movers component.
- Space Mono monospace font for all data values; Syne sans-serif for UI chrome.
- Interactive analog clock in the CTA section powered by vanilla JavaScript.
- Fully responsive metric cards, progress bars, and data tables.

6. Technical Stack & Dependencies

Library / Tool	Version	Role
Python	3.12.x	Core runtime. TensorFlow requires 3.12 (not 3.13+) for stable Keras integration.
Streamlit	Latest	Web application framework — renders the entire dashboard UI from Python code.
TensorFlow / Keras	Latest stable	Deep learning backend. Provides LSTM, Bidirectional, Dense, Dropout layers and training infrastructure.
yfinance	Latest	Yahoo Finance API wrapper — downloads OHLCV data for any ticker globally.
pandas	Latest	DataFrame operations: data loading, feature engineering, slicing, MultiIndex handling.
NumPy	Latest	Numerical arrays, sequence construction, RMSE/MAPE calculations.
scikit-learn	Latest	MinMaxScaler — scales features to [0,1] range for LSTM input normalization.
Matplotlib	Latest	Used in app.py prototype for training/prediction visualization plots.
Git	Any	Version control — tracked via .gitignore and .idea/vcs.xml IntelliJ configuration.

7. Development Roadmap & Version History

7.1 Version History

Version	Stage	Key Changes
v1.0	Prototype (app.py)	Single-stock (AAPL) LSTM with 1 feature (Close only), 1 epoch training, basic matplotlib plot output. Proved concept.
v2.0	Dashboard MVP	Streamlit frontend + dashboard backend split. User auth introduced. Basic LSTM 32→16 on crypto data.
v2.3	Bug Fix Release	Fixed yfinance MultiIndex column flattening crash. Fixed Series/DataFrame assignment error in BB_Width calculation. Fixed data leakage in validation.
v2.4 (Current)	Production Release	Bidirectional LSTM 128→64→32 architecture. 16 engineered features. 50-epoch + EarlyStopping training. 20-day leakage-free backtesting. Full dark/light theme UI.

7.2 Future Roadmap

The following enhancements are planned for subsequent releases:

- 17. Sentiment Analysis Integration: Incorporate Twitter/Reddit sentiment scores as additional LSTM features using transformer-based NLP models (FinBERT).
- 18. Portfolio Optimisation Module: Extend portfolio_data.csv analysis into a full Markowitz mean-variance optimiser with efficient frontier visualisation.
- 19. Production Authentication: Replace users.txt with a PostgreSQL database with bcrypt password hashing and session tokens.
- 20. Multi-Step Forecasting: Extend predictions from next-day to 7-day and 30-day rolling forecasts using encoder-decoder LSTM architecture.
- 21. Live Price Streaming: Replace periodic yfinance refresh with WebSocket-based real-time price feeds from Binance or CoinGecko APIs.
- 22. Model Persistence: Save trained model weights to disk (HDF5/SavedModel format) to eliminate retraining on every prediction request.
- 23. Docker Containerisation: Package the full stack in a Docker image for one-command deployment on any cloud provider.

8. Model Evaluation & Results

8.1 Performance Metrics

NeuralQuant evaluates model quality using two primary metrics:

- **RMSE (Root Mean Squared Error):** Used during training in `app.py` for the AAPL prototype. Penalises large errors quadratically, making it sensitive to outlier predictions.
- **MAPE (Mean Absolute Percentage Error):** Used in the production dashboard for backtesting. Expresses prediction error as a percentage of actual price, making results interpretable and comparable across different coins and price scales.

8.2 Observed Results from Prediction History

Analysis of the recorded `jutt_prediction_history.txt` file reveals the following performance data from May 2026 live predictions:

Date	Coin	Predicted (\$)	Actual (\$)	Abs. Error (%)
2026-05-07 21:02	ETH	2,357.30	2,292.98	2.80%
2026-05-07 21:37	BTC	80,354.67	80,119.99	0.29%
2026-05-08 11:22	BTC	78,465.05	80,556.61	2.60%
2026-05-12 16:19	BTC	80,513.51	80,754.97	0.30%
2026-05-12 16:35	BTC	81,406.70	80,754.97	0.81%
2026-05-12 17:04	BTC	81,488.76	80,379.18	1.38%

Average observed absolute error across all 6 live predictions: approximately 1.36% — well within the 5% 'High Accuracy' threshold. BTC predictions consistently achieved sub-1% error on same-day re-runs, demonstrating model stability.

9. Installation & Deployment Guide

9.1 Prerequisites

- Python 3.12.x (mandatory — TensorFlow is incompatible with Python 3.13+)
- Node.js / npm (optional, for documentation generation)
- Git
- VS Code or PyCharm (recommended IDE)

9.2 Step-by-Step Installation

24. Clone the repository: `git clone https://github.com/your-username/NeuralQuant.git`
25. Navigate to the project directory: `cd NeuralQuant`
26. Create a Python 3.12 virtual environment: `python -m venv venv`
27. Activate the virtual environment: `.\venv\Scripts\activate` (Windows) or `source venv/bin/activate` (macOS/Linux)
28. Install all dependencies: `pip install streamlit pandas numpy yfinance scikit-learn tensorflow keras matplotlib`
29. Launch the dashboard: `streamlit run frontend.py`
30. Open browser at `http://localhost:8501` and register a new account.

9.3 File Requirements

Ensure the following files are present in the working directory before launching:

- `CryptocurrencyData.csv` — master coin directory (4,150 rows)
- `portfolio_data.csv` — historical portfolio data
- `dashboard.py` — backend logic module
- `frontend.py` — Streamlit application entry point

10. Risk Assessment & Disclaimers

10.1 Technical Risks

- **yfinance API Dependency:** Yahoo Finance has historically changed its API structure without notice, requiring periodic fixes to the `fetch_live()` function (as occurred with the `MultilIndex` column issue fixed in v2.3).
- **Model Overfitting:** Despite Dropout layers and EarlyStopping, LSTM models can overfit to specific market regimes. Performance degrades significantly during unprecedented market events (black swans).
- **Computational Cost:** Training a Bidirectional LSTM on 4+ years of data with 16 features on CPU can take 2–8 minutes per prediction. GPU acceleration (CUDA) is recommended for production use.
- **Data Quality:** `CryptocurrencyData.csv` is a static snapshot. Prices, rankings, and market caps become stale immediately. The live prediction pipeline uses yfinance for actual model data, but the coin directory UI reflects snapshot values.

10.2 Financial Disclaimer

IMPORTANT DISCLAIMER

NeuralQuant is an academic and research tool. All predictions generated by the LSTM model are experimental outputs of a statistical model and DO NOT constitute financial advice, investment recommendations, or solicitations to buy or sell any cryptocurrency. Past model

accuracy does not guarantee future performance. Cryptocurrency markets are highly volatile. Always conduct independent research and consult a qualified financial advisor before making investment decisions.

11. Conclusion

NeuralQuant represents a mature, production-quality implementation of deep learning applied to the cryptocurrency prediction domain. Starting from a simple AAPL single-feature LSTM prototype (app.py), the project has evolved through iterative engineering into a full-featured, authenticated, multi-coin prediction platform with:

- A 3-layer Bidirectional LSTM architecture processing 16 technical features over 60-day lookback windows.
- Rigorous anti-leakage backtesting delivering honest MAPE scores on truly out-of-sample data.
- A professionally designed, responsive Streamlit dashboard supporting dark/light themes, animated UI components, and persistent user prediction histories.
- Live observed prediction accuracy of ~1.36% mean absolute error across 6 real BTC/ETH predictions in May 2026.

The platform stands as both a functional trading intelligence tool and a well-structured educational resource for understanding the full pipeline from raw financial data to deployed LSTM predictions. The modular architecture (dashboard.py / frontend.py separation) makes it straightforward to extend with new models, data sources, or UI components.

The roadmap items outlined in Section 7.2 — particularly sentiment analysis integration, model persistence, and multi-step forecasting — position NeuralQuant for continued development into a genuinely competitive retail crypto analytics product.

 **NeuralQuant — Predict Before the Market Moves**

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